
Software Engineering HS26

Project Proposal

"Forge Synesis"

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1 Introduction

This project introduces a study-oriented extension for the academic reference management platform JabRef. The extension is designed to support students in processing and studying learning material by transforming documents such as lecture scripts, notes, and other course-related sources into organized and interactive flashcards.

The core functionality uses an AI-based approach to generate flashcards from user-provided learning material. Users will be able to configure parameters such as the number and difficulty of the generated flashcards and subsequently review, modify, and study the generated content. This allows learning material that is already managed within JabRef to be transformed into an interactive study resource without requiring a separate application.

In addition to the flashcard functionality, the extension will incorporate a gamification system based on experience points and progression. Users can receive points for completing learning activities and track their progress over time. The purpose of this system is to provide additional motivation and encourage regular engagement with the learning material.

By combining JabRef's existing document and reference management capabilities with AI-supported flashcard generation and gamification, the project aims to create an integrated workflow for organizing, processing, and studying academic learning material.

2 Main Goal

The main goal of this project is to design and implement a study tool integrated into JabRef that helps students understand, practice, and retain learning material from university courses. The tool will use artificial intelligence to transform existing learning material, such as lecture scripts, notes, and other documents, into interactive learning resources.

The first implementation will focus on AI-generated flashcards. Users will be able to provide learning material and specify parameters such as the desired number of flashcards and their difficulty. The material will then be processed using the Gemini API, which will generate questions and answers based on the concepts and information contained in the provided sources.

The generated flashcards will be organized according to their respective subject or learning material. This allows users to build a dedicated collection of learning resources for each course and to review the material directly within the study tool.

A central objective is to ensure that the generated learning resources remain closely connected to the original source material. The AI should therefore generate questions based primarily on the content provided by the user rather than introducing unrelated information. The user should also be able to review and modify generated flashcards before using them for studying.

The resulting functionality should provide a simple workflow:

Import learning material → Generate flashcards → Review and edit → Study → Track progress

The flashcard functionality will serve as the foundation for additional learning features that may be implemented during the project.

3 Secondary Goal

The secondary goal of the project is to introduce gamification mechanisms that encourage users to study regularly and engage with the learning material.

The planned gamification system will reward users for completing learning activities. For example, completing or correctly answering flashcards can provide experience points (XP). Accumulated experience points can contribute to a user's level and potentially unlock achievements or other rewards.

The purpose of the gamification system is not to replace effective learning methods, but to provide additional motivation and feedback. Users should be able to see their progress over time and receive a sense of achievement from consistently studying their material.

Possible elements of the gamification system include:

- Experience points for completed learning activities
- User levels based on accumulated experience
- Achievements for reaching specific milestones
- Progress statistics
- Study streaks or other indicators of consistency

The exact scope of the gamification system will depend on the available development time and the complexity of the core flashcard functionality. The flashcard generation and studying workflow will remain the primary focus of the project.

4 Functional Requirements

The functional requirements describe the features and operations that the study extension must provide to the user. The primary focus is the generation, management, and use of AI-generated flashcards based on user-provided learning material.

- The system shall allow users to select or provide learning material that can be used as the basis for generating flashcards. This material may include lecture scripts, notes, PDFs, or other documents that are accessible through JabRef.
- The system shall provide users with configuration options for flashcard generation. At minimum, users shall be able to specify the desired number and difficulty of the generated flashcards. Depending on the final implementation, additional parameters such as language, topics, or question types may also be provided.
- The system shall use the Gemini API to generate flashcards based on the selected learning material and the parameters provided by the user. The generated flashcards shall contain a question or prompt and a corresponding answer, based on the material provided.
- After generation, the system shall display the flashcards to the user and allow them to review the generated content and correct it.

- The system shall provide a study mode in which users can actively review their flashcards. During a study session, the user shall be able to view the question or prompt, reveal the corresponding answer, and proceed through the selected set of flashcards.
- The extension shall organize generated flashcards according to their associated learning material or subject into collections for separate courses.
- In addition to the core flashcard functionality, the system shall implement a gamification component. Users shall receive experience points for defined learning activities, such as completing flashcards or study sessions.
- Where implemented within the project scope, the system shall also provide achievements for predefined learning milestones. These achievements may reward activities such as completing a certain number of flashcards, maintaining regular study activity, or reaching a specific level.

5 Non-functional Requirements

The non-functional requirements define the quality characteristics, technical constraints, and operational conditions under which the study extension shall operate. Since the extension is integrated into JabRef, it should follow the existing architecture, development practices, and usability principles of the application.

- The extension shall maintain the responsiveness of the JabRef user interface during operations that may require significant processing time. In particular, communication with the external Gemini API shall happen asynchronously not block the main user interface.
- The system shall be reliable when interacting with external services. Network failures, unavailable API services, invalid API responses, and other communication errors shall be handled gracefully and not cause JabRef to crash.
- The extension shall be maintainable and follow the architectural and coding conventions of the existing JabRef project. New functionality should be separated into appropriate components with clearly defined responsibilities.
- The system shall be designed to support automated testing of its core functionality.
- Because learning material may contain personal, confidential, or otherwise sensitive information, the system shall clearly communicate to the user when data is transmitted to an external AI service. API credentials and other sensitive configuration data shall not be hardcoded into the source code.
- The extension shall be compatible with the JabRef version used as the development baseline for the project.
- The system shall provide a consistent and understandable user interface that fits into the existing JabRef application and not negatively affect the existing features.
- Finally, the architecture should allow the study functionality to be extended in the future. For example, additional AI providers, learning methods, or study-related features should be capable of being integrated without requiring a complete redesign of the existing system.

6 Technical Requirements

The project will be developed as an extension of the existing open-source JabRef application. The JabRef source code will be used as the foundation of the project, allowing the team to modify existing components and introduce new classes and functionality where required. The source code is publicly available through the official JabRef GitHub repository and will serve as the development baseline for the extension.

The extension will be developed using **Java**, in accordance with the programming language and technical environment used by JabRef. The implementation shall follow the existing architecture, coding conventions, and development guidelines of the JabRef project to ensure that the new functionality can be properly integrated into the existing application.

For AI-supported flashcard generation, the project will integrate the **Gemini API**. The extension will communicate with the API to provide selected learning material and user-defined generation parameters and will process the API response into flashcards that can be displayed and managed within JabRef. The implementation must also account for possible API errors, network failures, invalid responses, and other communication problems.

The project will be developed collaboratively using **version control**, allowing all team members to work on the codebase while maintaining a history of changes. Git and the project's GitHub repository will be used to coordinate development, manage branches and changes, and integrate the work of individual team members.

As a group software-engineering project, the development process will follow the software-engineering guidelines provided by the course. This includes applying appropriate software design principles, dividing the system into clearly defined components, documenting relevant decisions, testing the implemented functionality, and reviewing and integrating code changes systematically. The team will also use an organized development workflow to distribute tasks, track progress, and coordinate the implementation of the project.

7 Future Extensions

The project timeframe should allow the core functionalities as well as both the main and secondary goals to be implemented. Additional features, such as generating different types of learning cards, creating practice questions, or implementing a more comprehensive achievement system, may be developed as future extensions. These features will only be implemented if sufficient time remains after the core requirements and primary objectives have been completed.